INCH-POUND
MIL-DTL-16377/17C(SH)
26 May 2023
SUPERSEDING
MIL-DTL-16377/17B(SH)
30 June 2014

DETAIL SPECIFICATION SHEET

# FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR BERTHS, 8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 229.1, 231.1, 232.1, 229.1SSL, 231.1SSL, 232.1SSL, 229.1SSL-USB, 231.1SSL-USB, AND 232.1SSL-USB

This specification is approved for use by the Naval Sea Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

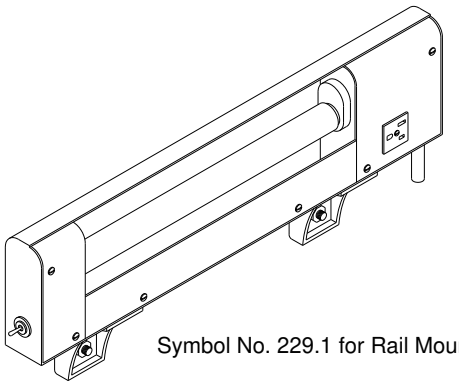
Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 9 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

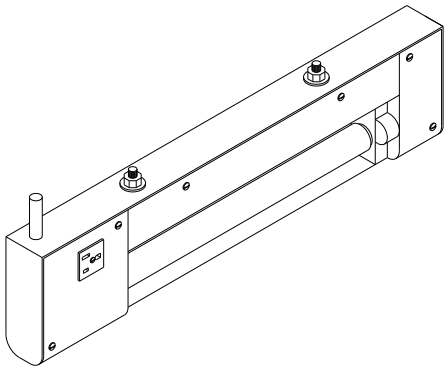
AMSC N/A FSC 6230

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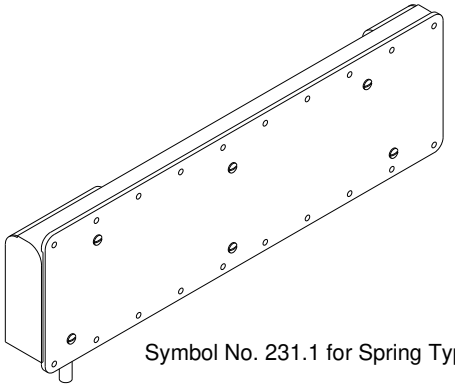
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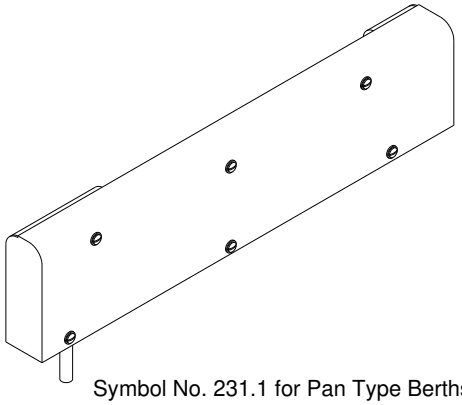
Symbol No. 229.1 for Rail Mounting



Symbol No. 232.1 for Bulkhead or Bar Mounting



Symbol No. 231.1 for Spring Type Berths



Symbol No. 231.1 for Pan Type Berths

FIGURE 1. Typical view of symbols 229.1, 231.1, and 232.1.



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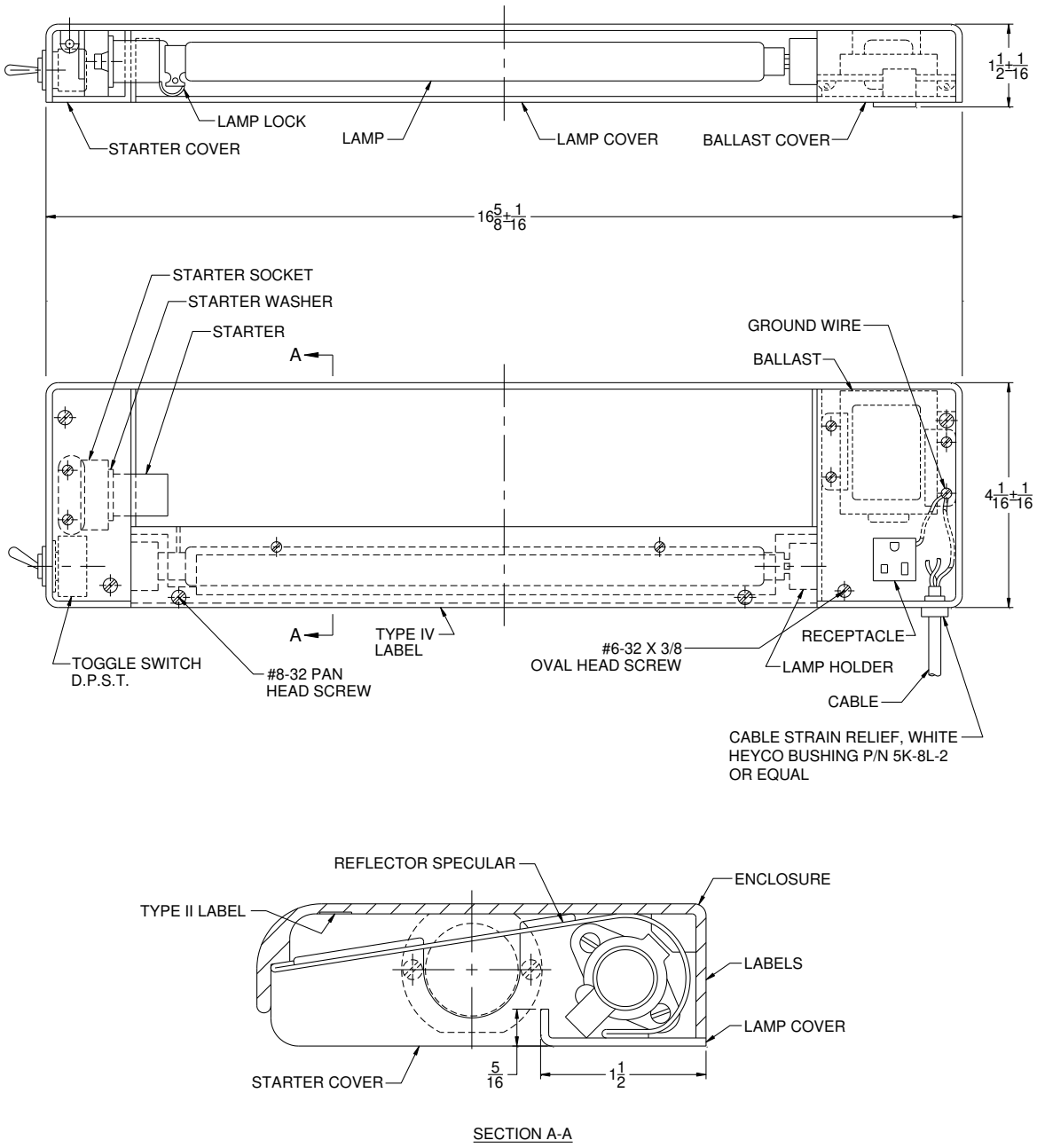



FIGURE 2. Typical plan and section views of basic light for symbols 229.1, 231.1, and 232.1.



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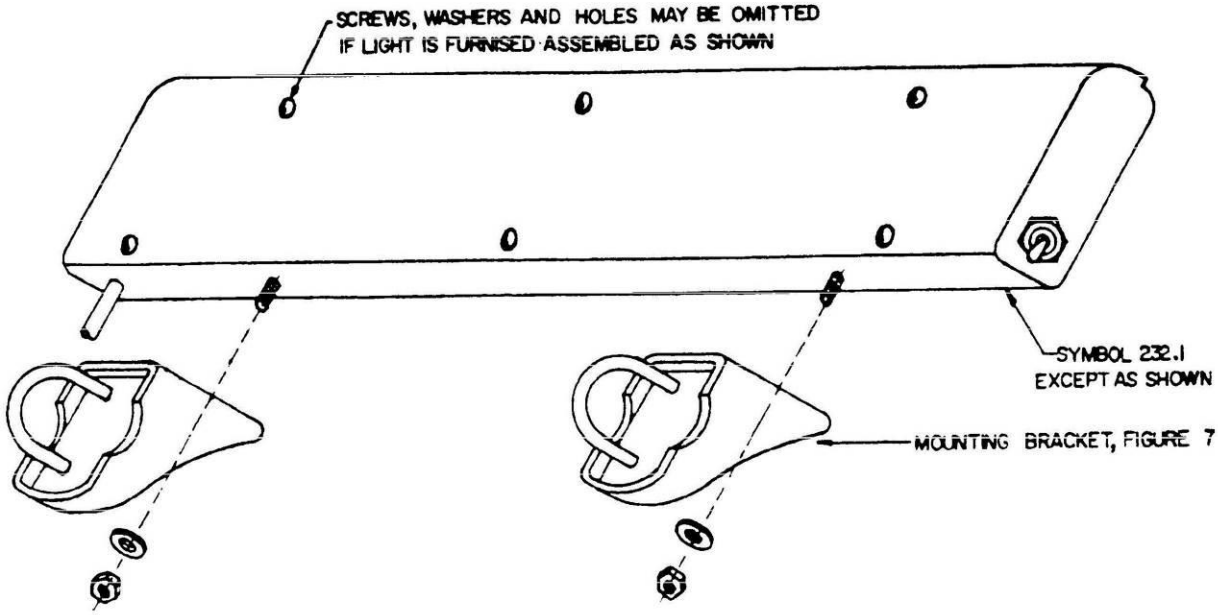



FIGURE 3. Typical view of symbol 229.1.

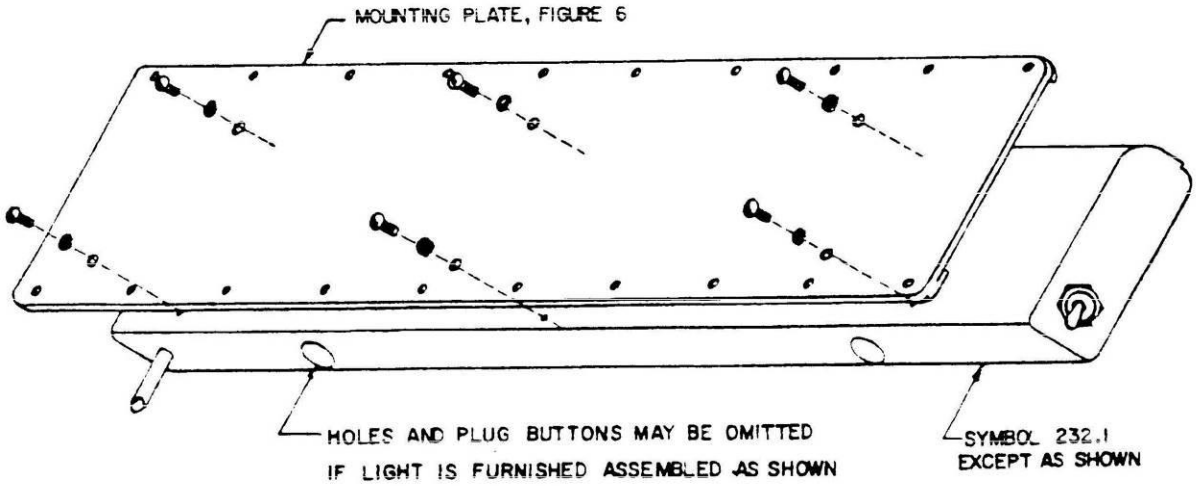



FIGURE 4. Typical view of symbol 231.1.



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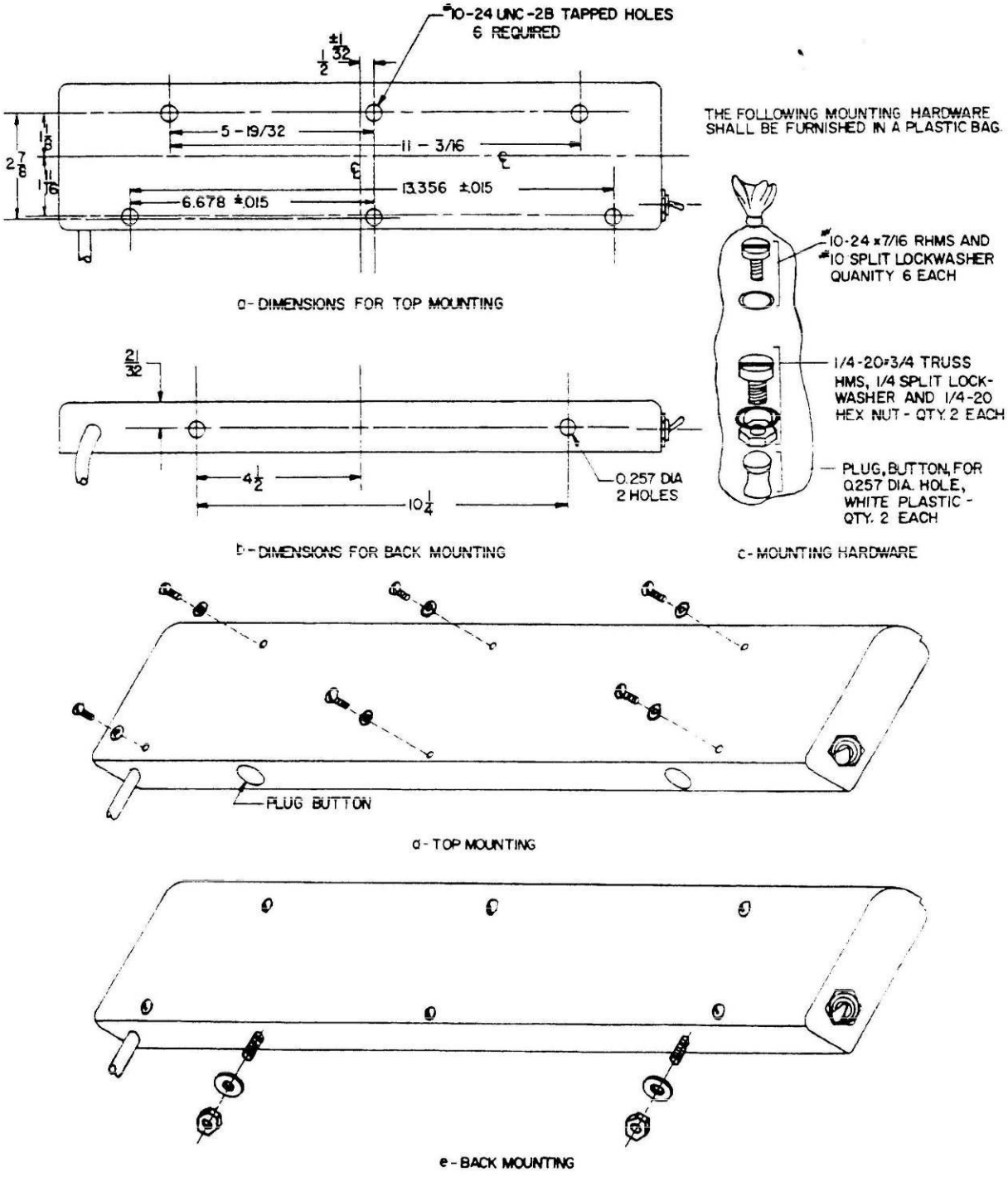



FIGURE 5. Typical view of symbol 232.1.



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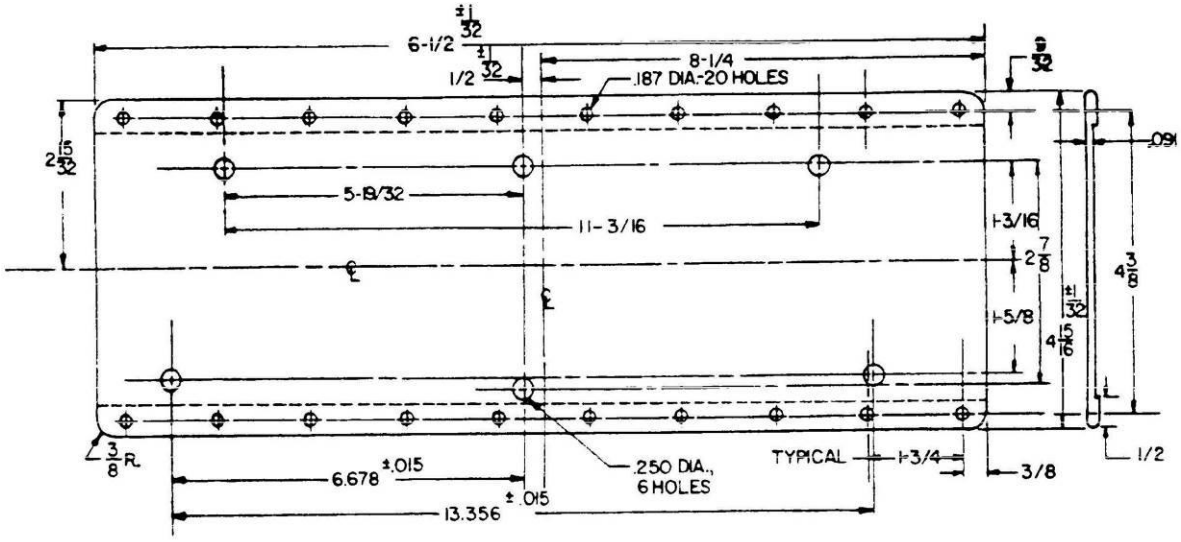



FIGURE 6. Mounting plate for symbol 231.1.

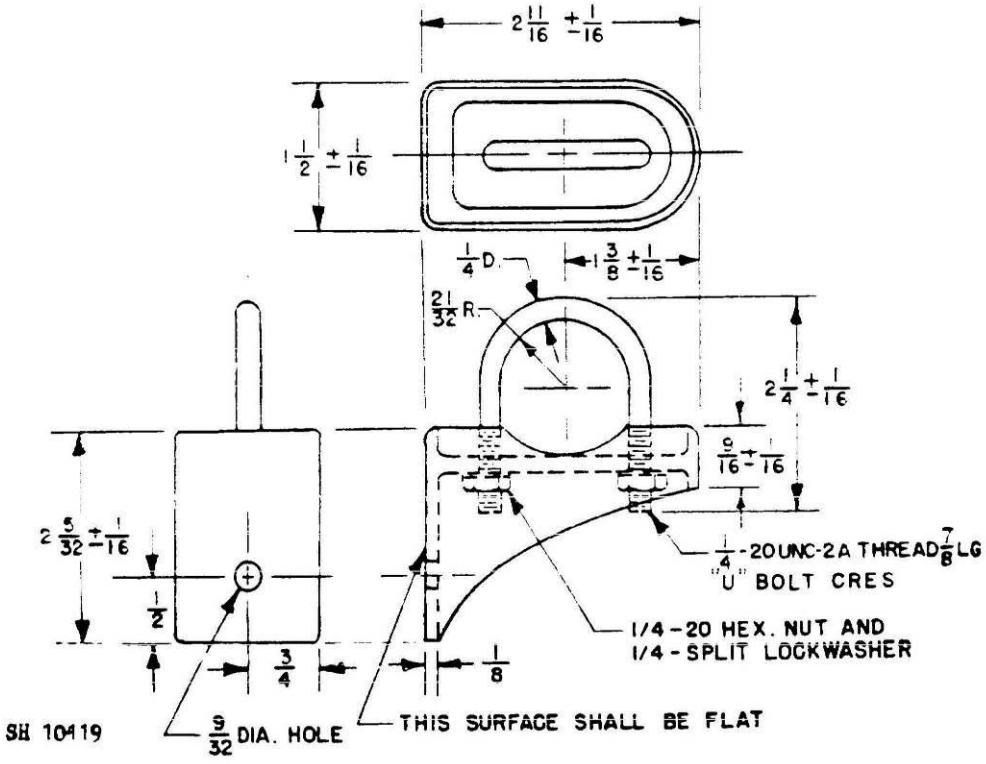



FIGURE 7. Mounting bracket for symbol 229.1.



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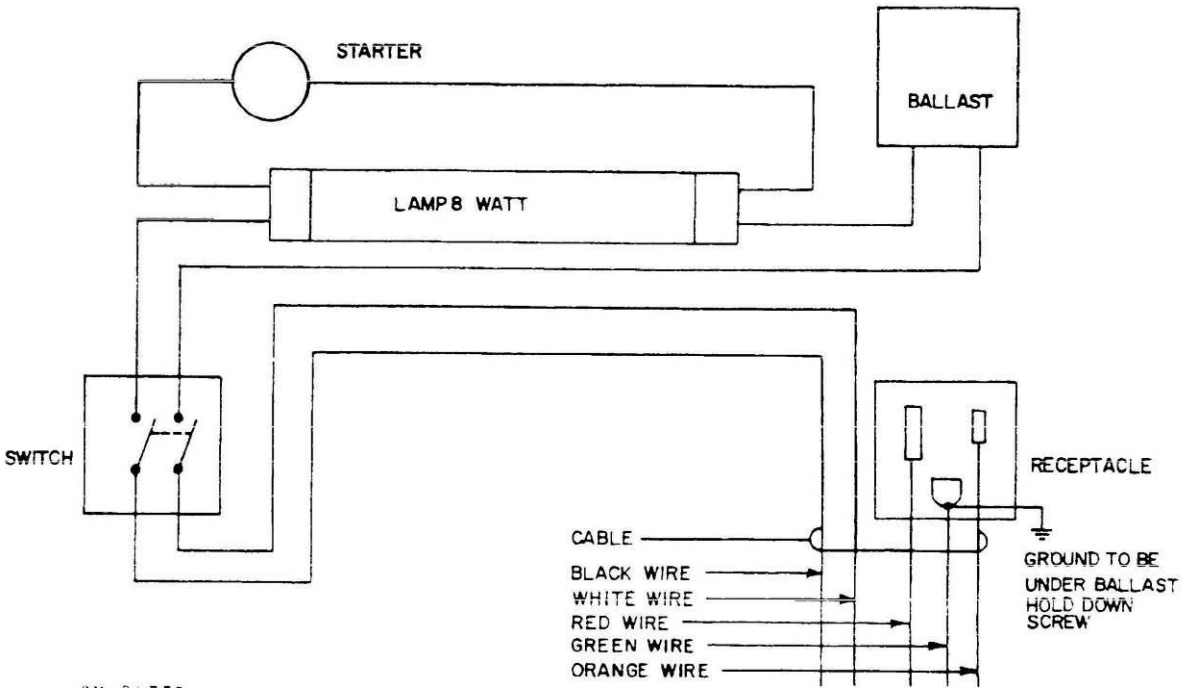
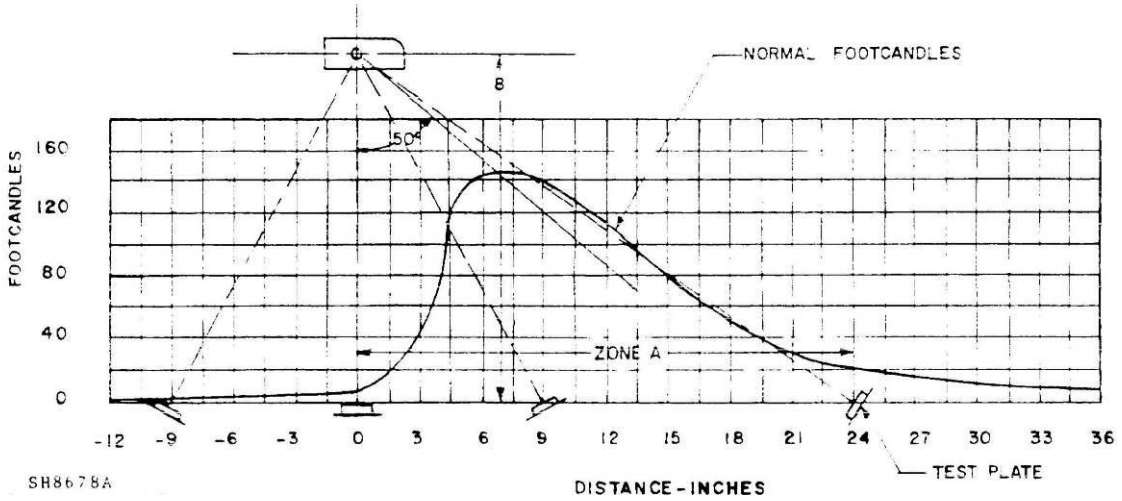



FIGURE 8. Wiring diagram.




Footcandle values within Zone “A” shall be not less than 90 percent of the value shown hereon. These results shall be met after correction to 380 lumens output for both lamp and ballast has been made.

Voltage: 115 volts, 60 hertz

Ambient temperature: 76 °F (24.4 °C)

Lamps: Lamp furnished with light shall be used during test

Ballast: Ballast furnished with light shall be used during test

Lamp correction: Distribution curves based on 380 lumens per lamp

FIGURE 9. Footcandle distribution curves.



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**FLUORESCENT REQUIREMENTS**

CLASSIFICATION

Type: I
Class: 1

SYMBOL NUMBERS

229.1: Complete lighting fixture assembly, for rail mounting, with one 8-watt cool white fluorescent lamp.

231.1: Complete lighting fixture assembly, for spring type berths (bunks), with one 8-watt cool white fluorescent lamp.

232.1: Complete lighting fixture assembly, for pan type berths, bulkhead or bar mounting, with one 8-watt cool white fluorescent lamp.

DESIGN

| | |
|---|---|
| Electrical rating: | 8 watts maximum, 120 volts, 60 hertz. |
| Dimensions and configurations: | See figures 1 through 7. |
| Photometric data: | See figure 9. |
| Lampholder spacing: | See figure 21 of MIL-DTL-16377. |
| Wiring diagram: | See figure 8. |
| Material: | Material for enclosure, lamp cover, mounting plate, and mounting brackets shall be aluminum or plastic type MAI-60 in accordance with ASTM D5948. |
| Finish: | Finish shall comply with 3.8.3.4 of MIL-DTL-16377, except that color shall be beige conforming to color number 26586 of SAE AMS-STD-595. |
| Cable strain relief: | Strain relief for cable shall be provided as shown on figure 2. |
| Grounding: | Aluminum enclosure: See figure 2;<br>Plastic enclosure: The reflector, ballast frame, and exterior metal parts of the switch shall be grounded to the receptacle ground contact. |
| Receptacle: | Receptacle shall be Leviton part number 1374-051 (ivory), or equal, except that the ground contact shall be nickel-plated, 0.0003 inch thick. |
| Cable: | 10 feet of type LSMHOF-7 in accordance with MIL-DTL-24643/7. |
| Weight: | 4 pounds, 2 ounces maximum. |
| Reflector: | Reflector shall be made of specular, pre-finished lighting sheet. |
| Designation and markings: | The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2.<br>(a) One type I.<br>(b) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I by name and stock number only. |
| Standard parts: | The standard parts shown in table I shall be used in the construction of the type I fixtures specified herein. |



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**SOLID-STATE LIGHTING (SSL) REQUIREMENTS**

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBER

| | |
|---|---|
| 229.1SSL: | Complete lighting fixture assembly, for rail mounting, with LED luminaire and white diffusing window assembly. |
| 231.1SSL: | Complete lighting fixture assembly, for spring-type berths (bunks), with LED luminaire and white diffusing window assembly. |
| 232.1SSL: | Complete lighting fixture assembly, for pan-type berths, bulkhead or bar mounting, with LED luminaire and white diffusing window assembly. |
| 229.1SSL-USB: | Complete lighting fixture assembly, for rail mounting, with LED luminaire and white diffusing window assembly with a Universal Serial Bus (USB) port. |
| 231.1SSL-USB: | Complete lighting fixture assembly, for spring-type berths (bunks), with LED luminaire and white diffusing window assembly with a USB port. |
| 232.1SSL-USB: | Complete lighting fixture assembly, for pan-type berths, bulkhead or bar mounting, with LED luminaire and white diffusing window assembly with a USB port. |

DESIGN

| | |
|---|---|
| Electrical rating: | 8 watts maximum (lighting load only), 120 volts, 60 hertz. |
| Dimensions and configuration: | See figures 1 through 7. |
| Photometric data: | Initial light output shall be within +20 percent and -10 percent of figure 9. |
| Chromaticity: | See 3.7.4 of MIL-DTL-16377. |
| Fail-safe circuit: | See 3.17 of MIL-DTL-16377. |
| Material: | Material for enclosure, lamp cover, mounting plate, and mounting brackets shall be aluminum or plastic type MAI-60 in accordance with ASTM D5948. |
| Finish: | Finish shall comply with 3.8.3.4 of MIL-DTL-16377, except that color shall be beige conforming to color number 26586 of SAE AMS-STD-595. |
| Cable strain relief: | Strain relief for cable shall be provided as shown on figure 2. |
| Grounding: | Aluminum enclosure: See figure 2;<br>Plastic enclosure: The reflector, ballast frame, and exterior metal parts of the switch shall be grounded to the receptacle ground contact. |
| Receptacle or USB port: | For the receptacle option, the receptacle shall be Leviton part number 1374-051 (ivory), or equal, except that the ground contact shall be nickel-plated, 0.0003 inch thick.<br>If specified by the acquisition activity, a USB port shall be installed instead of the receptacle. The USB port shall be easily accessible to devices that use the port. The port shall be a dedicated charging port device only, in accordance with Battery Charging Specification 1.2. The physical device shall be a USB type A connector. |
| Wiring: | If supplied with a receptacle, the fixture shall maintain the same control scheme for light output as the legacy fluorescent fixture. Refer to figure 8.<br>If the fixture is supplied with a USB port, the incoming power shall be supplied by two conductors of a three-conductor cable. Power to the USB port shall be supplied prior to the switch that controls the light. |



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| | |
|---|---|
| Cable: | For fixtures with receptacles, 10 feet of type LSMHOF-7 in accordance with MIL-DTL-24643/7.<br>For fixtures with USB ports, 10 feet of type LSTHOF-3 in accordance with MIL-DTL-24643/3. |
| Weight: | 4 pounds, 2 ounces maximum. |
| Designation and markings: | Two type I labels in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 3. |

## EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

| | |
|---|---|
| First article inspection: | |
| Complete fixtures: | See table II for type I and table III for type III. |
| Conformance inspection: | |
| Comparison inspection: | Same as first article inspection. |
| Conformance inspection: | See MIL-DTL-16377. |

## GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.

TABLE I. Standard parts.

| Name | Specification or Military Part Number | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| Lamp | ANSI/ANSLG C78.81 | 6240-00-299-5546 | Industry Number F8T5/CW |
| Lamplock | M16377/40-001 | 6250-00-399-7039 | --- |
| Lampholder | UL 496 | 6250-00-377-8590 | Supplement SA |
| Startersocket | UL 496 | 6250-00-295-2738 | Supplement SA |
| Starter | NEMA ANSI C78.180 | 6250-00-299-5962 | Commercial Designation: FS 5 |
| Ballast | M16377/45-001 | 6250-00-295-2735 | --- |
| Toggle switch, DPST | MS16569-1 | 5930-00-719-9659 | --- |



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TABLE II. First article and comparison inspection for complete type I fixtures, symbols 229.1, 231.1, and 232.1.

| Tests | Requirement Paragraph of MIL-DTL-16377 | Test Paragraph of MIL-DTL-16377 | Remarks |
|---|---|---|---|
| Examination | 3.12 | 4.5 | --- |
| Operation | 3.12 | 4.8.1 | --- |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Shock | 3.5.12 | 4.8.7 | 1 |
| Vibration | 3.5.13 | 4.8.8 | 1/ |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Noise | 3.5.14 | 4.8.9 | 1/ |
| Grounding circuit | 3.6.13.3 | 4.8.11 | --- |
| Light output | 3.7.2 | 4.8.6.2 | Footcandle |
| Magnetic permeability | 3.4.8 | 4.8.18 | --- |

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.



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TABLE III. First article and comparison inspection for complete type III fixtures, symbols 232.1SSL, 229.1SSL, 231.1SSL, 232.1SSL-USB, 229.1SSL-USB, and 231.1SSL-USB.

| Tests | Requirement Paragraph of MIL-DTL-16377 | Test Paragraph of MIL-DTL-16377 | Remarks |
|---|---|---|---|
| Examination | 3.12 | 4.5 | --- |
| Operation | 3.12 | 4.8.1 | --- |
| Power interface | 3.6.18 | 4.8.21 | --- |
| Electromagnetic interference | 3.6.14 | 4.8.16 | --- |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Shock | 3.5.12 | 4.8.7 | 1/ |
| Vibration | 3.5.13 | 4.8.8 | 1/ |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Noise | 3.5.14 | 4.8.9 | 1/ |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | --- |
| Photometrics | 3.7.1 | 4.8.6 | --- |
| Brightness | 3.7.5 | 4.8.6.4 | --- |
| Chromaticity | 3.7.4 | 4.8.6.5 | --- |
| Correlated color temperature, white light only | 3.7.4.1.1 | 4.8.6.6 | --- |
| Lumen maintenance | 3.7.11 | 4.8.20 | --- |
| Fail-safe circuit design | 3.17 | 4.8.24 | --- |
| Magnetic permeability | 3.4.8 | 4.8.18 | --- |

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.



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TABLE IV. Part number and general information.

| Military Part Number M16377/17- | Item Name | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| 229.1 | Complete light, symbol 229.1 | 6230-01-240-1105 | Figure 3 1/ |
| 231.1 | Complete light, symbol 231.1 | 6230-01-244-2936 | Figure 4 1/ |
| 232.1 | Complete light, symbol 232.1 | 6230-00-578-7548 | Figure 5 1/ |
| 232.1SSL | Complete fixture, solid state, receptacle option, symbol 232.1SSL | --- | --- |
| 229.1SSL | Complete fixture, solid state, receptacle option, symbol 229.1SSL | --- | --- |
| 231.1SSL | Complete fixture, solid state, receptacle option, symbol 231.1SSL | --- | --- |
| 232.1SSL-USB | Complete fixture, solid state, USB option, symbol 232.1SSL-USB | --- | --- |
| 229.1SSL-USB | Complete fixture, solid state, USB option, symbol 229.1SSL-USB | --- | --- |
| 231.1SSL-USB | Complete fixture, solid state, USB option, symbol 231.1SSL-USB | --- | --- |
| 001 | Mounting bracket for symbol 229.1 - two brackets | --- | Figure 7 1/ |
| 002 | Mounting plate for symbol 231.1 | --- | Figure 6 1/ |

NOTE:

1/ Former berth lights without receptacle (symbol 229, NSN 6230-00-578-7547; symbol 231, NSN 6230-00-808-4466; and symbol 232, NSN 6230-00-578-7548), procured for replacement purposes for ships where isolated receptacle circuits have not been provided for berth lights, shall be in accordance with symbols 229.1, 231.1, and 232.1, respectively, as specified herein except that the receptacle shall not be furnished and the cable shall be a three-conductor, LSTHOF-3, in accordance with MIL-DTL-24643/3.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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